

1 **Growth:** agent loops repeatedly ingest instructions, actions, and environment observations.

2 **Inference:** prefill, decode, dense attention, and KV memory create different bottlenecks.

3 **Architecture:** local, recurrent, sparse, and compressed mechanisms choose what history remains accessible.

4 **Caching:** stable prefixes avoid repeated prefill but do not reduce visible history.

5 **Compaction:** preserve intent, exact task state, recent work, and recovery paths—then evaluate continuation.